A Universal Multi-View Guided Network for Salient Object and Camouflaged Object Detection

Xiaogang Song, *Member, IEEE*, Pengfei Zhang, Xiaofeng Lu, Xinhong Hei, *Member, IEEE*, and Rongrong Liu

***Abstract*— Salient object detection and camouflaged object detection have attracted increasing attention due to their significant practical applications. While these two domains share similarities in recognition methods and object characteristics, they also exhibit distinctions. In this paper, we propose a novel multi-view guided network for camouflaged and salient object detection, utilizing the Transformer as the backbone network for feature extraction. Capitalizing on shared characteristics, we introduce a CNN-based multi-view encoder and a multi-view fusion module, enhancing the acquisition of multi-perspective information while minimizing the increase in computational cost. Moreover, recognizing domain differences, we incorporate an attention exploration module, seamlessly integrating multi-view features with globally extracted features from the backbone network. This integration involves simultaneous exploration from both positional and color perspectives, unearthing valuable information to identify salient and camouflaged objects. Our approach maximizes shared characteristics between the two tasks while effectively addressing their differences, leading to precise object identification—be it for camouflaged or salient objects. Extensive experiments on nine challenging benchmark datasets demonstrate the superior performance of our method across four widely used evaluation metrics, outperforming 34 state-of-the-art methods. Furthermore, we applied our method to other visually-related tasks, such as polyp segmentation and defect detection. The results further demonstrate the versatility of our model. The source code and results of our method are available at https://github.com/1900zpf/MVGNet.**

***Index Terms*— Salient object detection, camouflaged object detection, instance segmentation.**

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I. Introduction

GENERIC object detection(GOD) stands out as one of the most prevalent directions in the field of computer vision, serving as a cornerstone for numerous computer vision tasks. Further stratifying generic object detection is contingent upon the observer’s level of attention to the detected objects, leading to the delineation of salient object detection(SOD) and camouflaged object detection(COD). In this context, the goal of salient object detection is to segment the most visually striking objects in an image. The “salience” of an object may manifest in various aspects such as shape, size, color, and spatial location. Whether an object is deemed “salient” is often closely tied to its spatial position, with areas near the image edges generally not considered the most attention-grabbing regions in the entire image. Conversely, for camouflaged objects, spatial constraints are no longer limiting; they rely more on color camouflage to appear anywhere in the image and strive to seamlessly blend into the surroundings and background, making them difficult to perceive. We observe that these two seemingly contradictory types of objects have a certain conceptual overlap. As shown in Fig.1, we present a set of challenging animal photos where it’s difficult to definitively classify them as camouflaged or salient objects. From a spatial perspective, they exhibit features typical of salient objects, but from the standpoint of color variations, we can observe the camouflage strategies employed by these animals. This indicates that salient object detection and camouflaged object detection share a certain degree of correlation in detecting objects.

In addition to the correlation regarding the detected objects, we also observe a similarity in the detection methods employed by these two tasks. For instance, Sun et al. [1] utilize CNN as the backbone network to extract image features, capturing object boundary information as guidance for generating more accurate predictions of camouflaged objects. Wu et al. [2] propose a decomposition and completion framework, leveraging the backbone network and boundary information to jointly model the interior and boundaries of salient objects, thereby producing more comprehensive predictions for salient objects. Furthermore, Zhu et al. [3] design a texture label for camouflaged objects, refining multi-level texture cues to enhance segmentation accuracy. Similarly, Zhou et al. [4]. introduce texture information, refining the predicted saliency

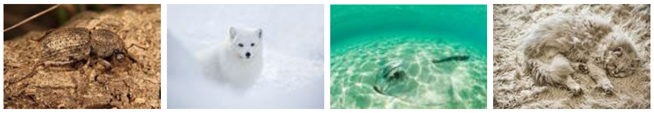

Fig. 1. The four images presented, when viewed from a color perspective, unmistakably fall within the category of camouflaged objects. However, from the standpoint of the spatial locations of the objects, they evidently all qualify as the most attention-grabbing "salient objects" within the entire image.

maps by matching texture information across different spatial contexts. It is evident that whether capturing boundary information as an aid or introducing texture labels to guide network predictions, these strategies not only significantly improve the accuracy of SOD tasks but also prove applicable to the field of COD. This, from another perspective, underscores the many commonalities existing between these seemingly contradictory tasks in both detection procedures and the detected objects.

Apart from this, we have also noted the significant role of the multi-view strategy in both tasks. Inspired by the human observation process, humans adjust their eye movements to gain different perspectives when recognizing targets of varying sizes. Specifically, for nearby targets, humans adopt a "widening eyes" strategy, while for distant targets, a "squinting eyes" strategy is employed. This is akin to adjusting the camera's focal length to obtain clearer images, and contrasting at different scales can assist in better discerning target categories. This approach is equally applicable to the detection of salient objects, where multi-view information can effectively enhance the model's capability to detect objects at different distances. For camouflaged object detection, the contrast at different viewpoints allows us to better discover those imperceptible color changes. Zheng et al. [5] proposed MFFN for detecting camouflaged objects, which generates multi-view information through data augmentation, and then explores and integrates this information to capture the boundaries and semantic information of camouflaged objects.

Building upon the aforementioned observations, this paper endeavors to devise a universal network architecture that caters to both tasks by capitalizing on their commonalities. Firstly, we employ the Transformer as the backbone network for feature extraction. Abundant research has demonstrated the irreplaceable advantages of the Transformer in capturing long-distance dependencies and extracting global information. The superior global perceptual capability is crucial for accurate localization in both salient and camouflaged object detection tasks. Secondly, we designed a dedicated convolutional branch for extracting multi-view information. Through experimentation, we observed that while multi-view information effectively enhances the model's recognition performance, it inevitably leads to an increase in computational complexity. To alleviate this issue, we innovatively designed an independent CNN branch for extracting and integrating multi-view information. Leveraging our carefully designed multi-view fusion module, we aimed to thoroughly exploit multi-view information while minimizing the additional computational load as much as possible. Thirdly, we introduce an innovative saliency detection module. This module processes the extracted features in parallel, exploring the saliency of objects from both color and positional perspectives. The attention scores learned through this process contribute to a more effective segmentation of objects—be they salient or camouflaged.

In summary, our contributions are as follows:

- We introduce a segmentation network designed for both salient and camouflaged objects. Our approach takes into full consideration the commonalities between the two while effectively addressing their differences, resulting in accurate prediction images.
- We have designed a multi-view feature extraction branch aimed at integrating multi-view feature information extracted from the CNN encoder. This branch not only enhances model segmentation performance but also effectively manages the increase in computational load. Additionally, we introduce an attention exploration module(AEM) that concurrently explores contextual information from both color and positional perspectives, generating precise prediction results.
- Extensive experiments on publicly available datasets for both salient and camouflaged objects demonstrate a significant improvement in our method compared to the state-of-the-art approaches. Additionally, we have applied our method to other related domains, such as polyp segmentation and defect detection, further confirming the strong generality of our approach.

II. Related Work

A. Salient Object Detection

In recent years, salient object detection has remained a prominent research topic in the field of computer vision. This is primarily attributed to its significant application value in various downstream tasks, including object detection [6], visual tracking [7], scene classification [8], medical image segmentation [9], [10], semantic segmentation [11], [12], [13], and applications in virtual reality (VR) and augmented reality(AR) [14].

Simultaneously, numerous excellent salient object detection models have emerged. From the perspective of the backbone network, they can be roughly categorized into two classes: 1) CNN-base. He et al. [15] proposed a super pixelwise convolutional neural network, which learns the internal features of image saliency by restoring contextual information. Li et al. [16] introduced an improved method to enhance the spatial consistency of salient objects. They further utilized multi-scale feature aggregation to generate salient object segmentation maps. Wang et al. [17] presented a neural network based on the Fast R-CNN framework. This network integrates boundary information and multi-scale contextual information, aiming to segment salient objects by expanding the network's contextual awareness. 2) Transformer-base. Liu et al. [18] introduced a unified model based on the Transformer architecture for RGB and RGB-D salient object detection. This model utilizes the modeling capabilities of the Transformer to accurately predict the saliency of images by capturing long-range dependency relationships. In comparison to CNN architectures, it has achieved an exciting performance enhancement. Yun and Lin [19] proposed an end-to-end salient object detection network based on the Transformer architecture. They devised a local context branch to assist the backbone in feature learning, aiming to achieve a more comprehensive structure and clearer detailed information in predictions. Li et al. [20] proposed a novel Dense Attention Feature Enhancement Module for efficient feature enhancement in saliency detection. This module can be conveniently inserted into other saliency detection models to improve their performance. Song et al. [21] proposed a dual-branch encoder for feature extraction. The backbone network employs the Transformer to capture saliency information, while the auxiliary branch utilizes CNN to refine fine details. Additionally, boundary information is introduced for joint fusion, resulting in the salient object prediction map.

B. Camouflaged Object Detection

Camouflaged object detection is dedicated to segmenting objects that seamlessly blend into the background. It is more challenging compared to the task of salient object detection. However, concurrently, due to its capability to identify camouflaged objects, COD models have significant applications in various domains, such as polyp segmentation [22], [23], species discovery [24], retinal imaging segmentation [25] and agriculture [26], [27].

Camouflage object detection algorithms, in terms of the choice of backbone network, can be categorized into two types, similar to SOD: CNN and Transformer. Given that the segmentation of camouflaged objects often draws inspiration from human discernment of such objects, from this perspective, we can classify them into two categories: 1) Boundary-Aided Method. Sun et al. [1], leveraging the feature information extracted by the backbone network, further explore valuable boundary information to guide the network in generating more accurate prediction maps. Chen et al. [28] proposed a boundary-guided fusion module designed to explore complementary relationships between camouflaged regions and object boundaries, thereby further refining the detailed information in predictions. Zhu et al. [29] devised a separated attention mechanism to highlight the boundary information of camouflaged objects. The obtained results are further integrated to enhance the perceptual ability for boundaries, thereby improving the accuracy of camouflaged object recognition. 2) Building upon the boundary, Zhu et al. [3] introduced additional texture labels. They designed a module that utilizes advanced features to guide the gradual fusion of low-level features, aimed at enhancing segmentation accuracy. Similarly recognizing the role of texture information, Ji et al. [30] designed a texture encoder for the network to extract texture features. Through a soft grouping approach, they comprehensively fused texture information to obtain accurate prediction maps.

III. METHOD

In this section, we elaborate on the proposed network framework. In Sec.III-A, we provide an overview of the complete network architecture. In Sec.III-B, we introduce the proposed multi-view encoder and multi-view fusion module. In Sec.III-C, we present the proposed attention exploration module. Finally, in Sec.III-D, we describe the loss functions employed in this paper.

A. Network Structure

The overall pipeline of the proposed MVGNet is illustrated in Fig.2, which is an end-to-end trainable framework.

In this paper, we employ the Pyramid Pooling Transformer network (P2T [31]) as our backbone network for feature extraction. The input image$I \in \mathbb{R}^{H\times W\times 3}$ is in RGB 4:4:4 format with channel 3, height H and width W. As shown in the orange section in Fig.2, After undergoing preprocessing operations, the input is fed into the encoder, ultimately yielding four sets of feature maps$\left\{t_i \in \mathbb{R}^{\frac{H}{2^{i+1}}\times\frac{W}{2^{i+1}}\times C_i}, i = 1, \ldots, 4\right\}$. The reason for selecting P2T as the feature extractor lies in its adoption of pyramid pooling strategy, which effectively reduces the computational burden compared to traditional Transformers. This makes it more suitable for visual tasks, particularly well-suited for computationally demanding dense prediction tasks such as image segmentation. In this paper, the four sets of global features extracted by the backbone network undergo channel adjustment, with the number of channels adjusted to 64, denoted as C.

Additionally, we have devised a multi-view encoder to extract multi-view information. This encoder, implemented based on CNN, is designed to extract valuable feature information from multi-view input images at the expense of minimizing computational costs. As indicated by the green section in Fig.2, the input to this encoder is the processed multi-view images, including the original image $I^{1.0x}$, the enlarged original image $I^{1.5x}$, and the reduced original image $I^{0.5x}$. This configuration is designed with the consideration that certain objects are better suited for an enlargement strategy to improve recognition accuracy, while others are more easily discernible in the reduced perspective. By inputting these three perspectives into the encoder and leveraging our encoder and multi-view fusion module, we assign different weight ratios to filter out perspective information that is more conducive to detection. Further details about this module will be presented in Sec.III-B, ultimately yielding three sets of features denoted as $\left\{C_i^{1.5x}, C_i^{1.0x}, C_i^{0.5x}, i = 1, \ldots, 4\right\}$. Subsequently, our multi-view encoder will reexplore these three sets of features, learning the connections and distinctions between each channel, attenuating or eliminating ineffective information, amplifying the feature weights conducive to distinguishing the target, ultimately yielding a processed feature set denoted as f_5 and a prediction map p_5.

As shown in the blue section of Fig.2, our core module is the saliency exploration module. Our network is a unified framework applicable to both SOD and COD. The scaling strategy extracts and integrates multi-perspective information, benefiting the recognition of both tasks. The saliency exploration module is designed with consideration of the differences between the two tasks. It consists of two parallel parts exploring color and spatial positions. The color exploration part is inclined to discover objects hidden in the background, while the position exploration part is inclined to identify salient

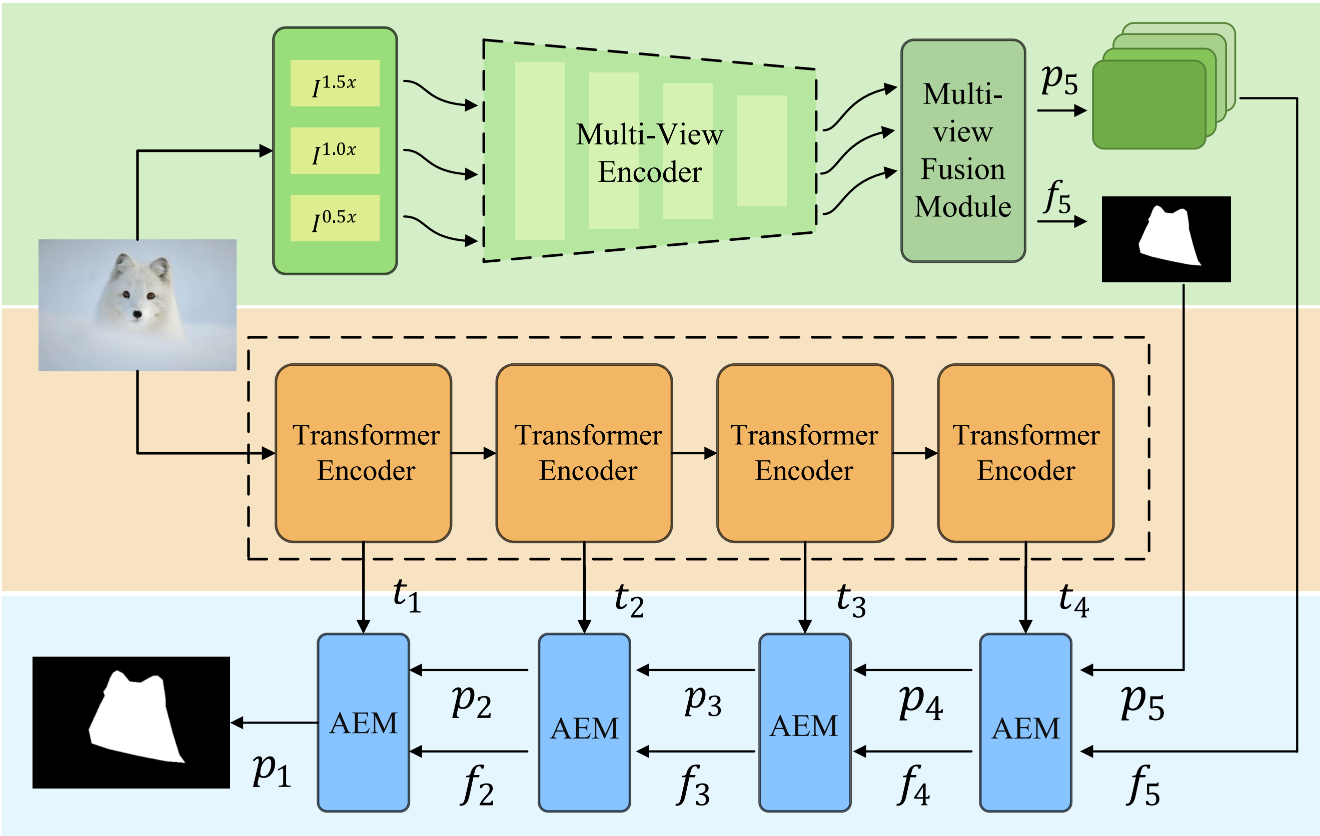

Fig. 2. The overall pipeline of the proposed MVGNet. We employed a Transformer as the backbone network for feature extraction. Additionally, we devised a CNN-based multi-view encoder to extract multi-view features. The fused multi-view features, along with the globally extracted features from the Transformer, were fed into our designed attention exploration module(AEM). This module systematically explores features from both the camouflaged and salient perspectives, ultimately producing high-resolution prediction images.

objects in the image. However, this is not absolute. Therefore, the effective integration of the exploration results from both parts is crucial for accurately recognizing objects. Simultaneously, we incorporate higher-level features and prediction maps in the recognition process, progressively restoring high-resolution prediction maps from small to large to obtain more comprehensive details. Therefore, the overall design of our network is illustrated in Fig.2. Starting from the commonalities in object detection, we simultaneously explore the differences between salient and camouflaged objects. The results are thoroughly fused to obtain accurate prediction images.

B. Multi-View Feature Extraction Branch

The multi-view feature extraction branch consists of a multi-view encoder and a multi-view fusion module(MFM). This branch aims to extract and integrate multi-view information, resulting in fused features and coarse prediction maps. The multi-view encoder is responsible for feature extraction, while the multi-view fusion module, as a generic component, does not focus on the saliency or camouflage properties of targets. Instead, it aims to discover objects in images as much as possible by integrating multi-view information, thereby enhancing the accuracy of both COD and SOD tasks.

The extraction and fusion of multi-view information often lead to a significant increase in computational complexity. To manage the consumption of computational resources,

TABLE I

DETAILS OF THE MULTI-VIEW ENCODER. k: KERNEL SIZE, c: OUTPUT CHANNELS, s: STRIDE, AND p: ZERO-PADDING

Layer	Input Size	Output Size	Component	k	c	s	p
1	$3 \times H \times W$	$64 \times \frac{H}{2} \times \frac{W}{2}$	ConvBR	7	64	2	3
2	$64 \times \frac{H}{2} \times \frac{W}{2}$	$64 \times \frac{H}{4} \times \frac{W}{4}$	ConvBR	3	64	2	1
3	$64 \times \frac{H}{4} \times \frac{W}{4}$	$64 \times \frac{H}{8} \times \frac{W}{8}$	ConvBR	3	64	2	1
4	$64 \times \frac{H}{8} \times \frac{W}{8}$	$64 \times \frac{H}{16} \times \frac{W}{16}$	ConvBR	3	64	2	1

we did not use the backbone network to extract multi-view information. Instead, we introduced a CNN-based multi-view encoder for this purpose. This approach effectively avoids an explosion in computational complexity. Simultaneously, a carefully designed fusion module is utilized to filter and enhance information from different perspectives, ensuring that we extract more beneficial multi-view information for object recognition. The detailed configuration information of the multi-view encoder is presented in Table I.

The multi-view encoder serves as a shared encoder for extracting features from inputs of three different perspectives. After processing by the multi-view encoder, three sets of feature vectors are output, denoted as $\{C_i^{1.5x}, C_i^{1.0x}, C_i^{0.5x}, i = 1, \ldots, 4\}$. The result $C_4^{1.5x}, C_4^{1.0x}$, and $C_4^{0.5x}$, from the final layer of the encoder is used as the

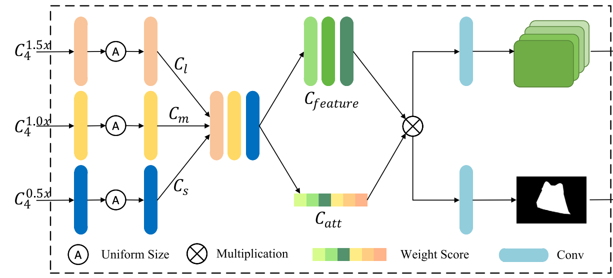

Fig. 3. Illustration of the Multi-view Fusion Module. Multiple-view images $C_4^{1.5x}$, $C_4^{1.0x}$, and $C_4^{0.5x}$ are taken as input, with their dimensions aligned first. Subsequently, the relationships between channels are explored to generate corresponding attention scores. Leveraging multiplication operations enhances effective features, ultimately obtaining filtered features along with the coarse prediction map.

input for MFM. The specific operation of MFM is illustrated in Fig.3. Initially, we unify the dimensions of the three sets of input features, resulting in C_l, C_m, and C_s, which correspond to the inputs $C_4^{1.5x}$, $C_4^{1.0x}$, and $C_4^{0.5x}$, respectively, as formulated below:

$$C_l, C_m, C_s = CBR(U(CBR(C_4^{1.5x}, C_4^{1.0x}, C_4^{0.5x}))) \quad (1)$$

where $CBR(\cdot)$ is the combination of convolution, batch normalization and $ReLU$. Unless otherwise specified, the convolutional operations in the $CBR(\cdot)$ module of this paper are configured with a kernel size of 3, a stride of 1, and padding of 1. $U(\cdot)$ denote bi-linear interpolation operations. The number of channels of the unified size features remains consistent with the input, at 64. Their size is $\frac{H}{32} \times \frac{W}{32}$, consistent with the output size of the fourth stage of the backbone network, facilitating subsequent computations. Subsequently, we employ the Cat operation to merge three sets of features. Then, the merged features are split into two parallel pathways. One pathway is responsible for exploring relationships between channels, producing channel attention scores of size 1×1 with 192 channels, denoted as C_{att}. The other pathway executes $CBR(\cdot)$ operations to obtain a set of features with unchanged size and 192 channels, denoted as $C_{feature}$. The formulations are as follows:

$$\begin{aligned} C_{att} &= Att(Cat(C_s, C_m, C_l)) \\ C_{feature} &= CBR(Cat(C_s, C_m, C_l)) \end{aligned} \quad (2)$$

where $Att(\cdot)$ denotes the channel attention mechanism, the specific operations contain pooling, convolution, $ReLU$ and $Softmax$. $CBR(\cdot)$ is the combination of convolution, batch normalization and $ReLU$. $Cat(\cdot)$ represents the operation of feature concatenation. Multiplying the attention scores with the feature groups weights the features from different perspectives.

Finally, using simple convolutional operations, the results are dimensionally reduced to 1 and 64, serving as the input features f_5 and the coarse prediction map p_5 for the next stage. The entire operation is represented by the following formula:

$$\begin{aligned} f_5 &= Conv_1(CBR(C_{feature} \times C_{att})) \\ p_5 &= Conv_2(CBR(C_{feature} \times C_{att})) \end{aligned} \quad (3)$$

where $Conv_1(\cdot)$ represents a convolution operation with an input channel of 192, an output channel of 64, and a kernel size of 3, and $Conv_2(\cdot)$ represents a convolution operation with an input channel of 192, an output channel of 1, and a kernel size of 3. $CBR(\cdot)$ is the combination of convolution, batch normalization and $ReLU$. The final output consists of the fused feature map f_5 and the coarse prediction map p_5 with a single channel.

C. Attention Exploration Module

The multi-view feature extraction branch focuses on the commonality of SOD and COD tasks, which is to segment objects in images. Meanwhile, the attention exploration module aims to explore the differences between these two tasks. Specifically, this module takes global features and multi-view information as input and consists of two parts: color exploration and position exploration. The color exploration part delves into the relationships between channels to discover camouflaged objects, while the position exploration part concentrates on spatial variations to detect salient objects. Finally, the information from both branches is fused as input for the next stage. The overall structure is shown in Fig.4.

For the color exploration component, we first adjust the size of the input feature f_i to obtain the query Q_i, key K_i and value V_i, respectively, where $Q_i, K_i, V_i \in \mathbb{R}^{C \times N}$ and $N = H_i \times W_i$ is the number of pixels. Next, we reshape the coarse prediction map p generated in the previous stage and fuse it with Q_i and K_i through multiplication. The transposition of the fused K_i is then matrix-multiplied with the fused Q_i, followed by applying softmax to calculate the channel attention map. To distinguish it from the position exploration component, we denote the result as $f_c' \in \mathbb{R}^{C \times C}$:

$$f_c' = (Q_i \times reshape(p_i)) \times (K_i \times reshape(p_i))^T \quad (4)$$

Perform matrix multiplication between f_c' and value V_i to reshape the aggregated attention features to $\mathbb{R}^{C \times H \times W}$. Finally, apply a residual connection with the input f_i to obtain f_c''.

For the position exploration component, corresponding $\widehat{Q_i}$, $\widehat{K_i}$, and $\widehat{V_i}$ are generated. However, differently, a 1×1 convolution is applied first to perform channel dimension reduction on the input feature before adjusting its size to generate $\widehat{Q_i}$ and $\widehat{K_i}$. The number of channels is reduced from C to $C/8$. The upper-level prediction map is introduced using multiplication, and then the two are subjected to matrix multiplication to obtain $f_p' \in \mathbb{R}^{N \times N}$, $N = H_i \times W_i$.

$$f_p' = (\widehat{Q_i} \times reshape(p_i)) \times (\widehat{K_i} \times reshape(p_i))^T \quad (5)$$

Similar to the other part, f_p' is aggregated with value $\widehat{V_i}$, and then subjected to a residual connection to obtain f_p''. These two parts run in parallel, and in the end, we need to fuse the results of exploration, f_c'' and f_p''. Additionally, we introduce the high-level features t_{i-1} extracted by the Transformer to gradually integrate and generate a high-resolution prediction map:

$$d_{i-1} = Cat(CBR(f_c'' \times t_{i-1}), CBR(f_p'' \times t_{i-1})) \quad (6)$$

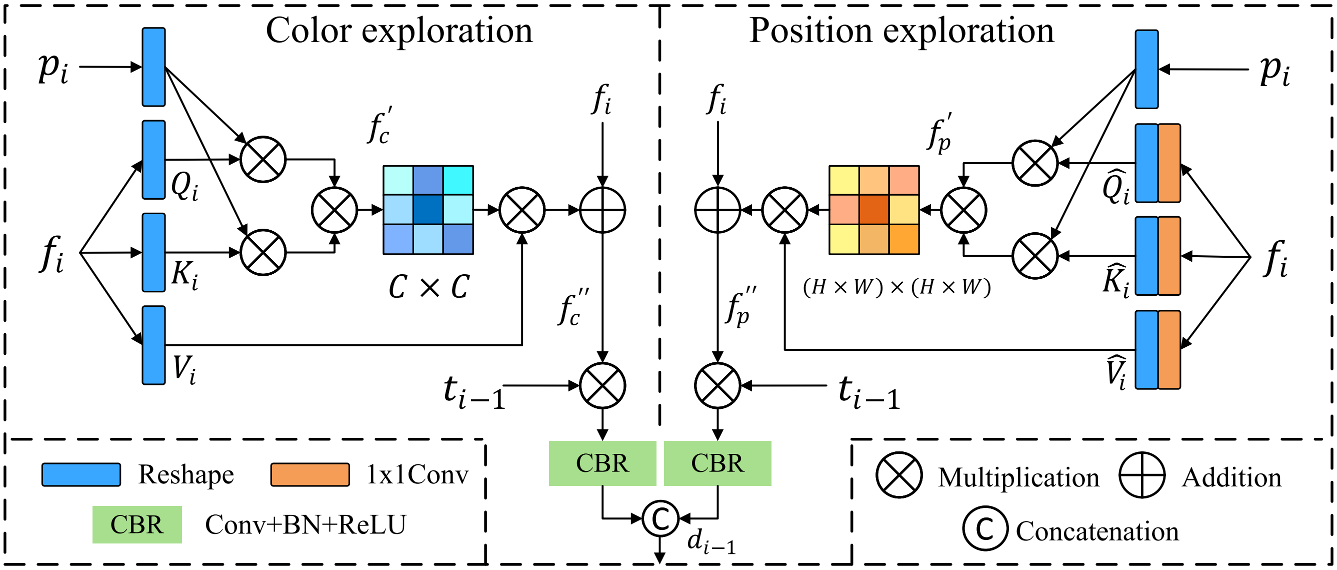

Fig. 4. Illustration of the Attention Exploration Module. The module is divided into two parallel executions. The color exploration part is responsible for exploring color relationships in the image from a channel perspective to identify camouflaged objects. The position exploration part is responsible for exploring the spatial relationships of targets from a spatial perspective to identify salient objects. Finally, the exploration results from both parts are thoroughly integrated.

Utilizing d_{i-1} to generate features f_{i-1} and the prediction map p_{i-1}, contributing to the computations in the next stage.

$$f_{i-1} = CBR(CBR(d_{i-1}) + t_{i-1})$$
$$p_{i-1} = CBR(CBR(d_{i-1}) + t_{i-1}) \tag{7}$$

This module is employed four times, accomplishing the reconstruction of prediction maps from coarse to fine. Throughout this process, careful consideration is given to the distinctions between SOD and COD tasks, and the globally extracted features from the Transformer are ingeniously integrated, rendering it applicable to detection tasks in diverse scenarios.

D. Loss Function

In the field of image segmentation, a common practice involves employing a hybrid loss function to supervise the network training process, comprising weighted intersection-over-union($wIoU$) loss and weighted binary cross-entropy($wBCE$) loss [32]. In this paper, to enhance the effective supervision of object boundaries, an uncertainty-aware loss(UAL) [33] is introduced. This function primarily emphasizes uncertainty and fuzziness in the predicted results, thereby better guiding the network in recognizing detailed information about objects.

$$L_{total} = L_{bce}^{w} + L_{IoU}^{w} + L_U \tag{8}$$

where L_U represents the uncertainty-aware loss function, and the specific formula is expressed as follows:

$$L_U = \lambda \ \times (1 - |2p - 1|^2) \tag{9}$$

where λ is the balance coefficient and used the cosine strategy to set λ. p is the value of pixels in the image. Additionally, besides supervising the final results, we also applied the same supervision to the coarse prediction results of the CNN branch. Therefore, the supervision strategy of our network during the training process is as follows:

$$L = L_{total}(P, G) + L_{total}(P_{cnn}, G) \tag{10}$$

where P and G represent the predicted map and ground-truth label, respectively. P_{cnn} represents the prediction map of the CNN branch.

IV. Experiment

A. Implementation Details

We constructed our model using the PyTorch library [34], employing P2T [31] as the backbone network pre-trained on the ImageNet dataset [35]. The complete model underwent training for 50 epochs on an NVIDIA 3090Ti GPU to achieve optimal results. Throughout training, input images were uniformly resized to 352×352. The learning rate was initialized at 0.0001 and followed the cosine preheat decay strategy. The optimizer utilized was SGD with a momentum of 0.9 and a weight decay of 0.0005.

B. Datasets and Evaluation Metrics

As the proposed network in this paper is a unified framework applicable to both SOD and COD, experiments were conducted on separate publicly available datasets for each task to validate the performance of our network.

For the SOD task, we evaluated our approach on five widely used public datasets: DUTS [36], ECSSD [37], HKU-IS [16], PASCAL-S [38], and DUT-OMRON [39]. Specifically, ECSSD comprises 1000 images containing salient objects. HKU-IS consists of 4447 images with multiple foreground objects. OMRON comprises 5168 images, each containing at least one object. PASCAL-S is constructed from a semantic segmentation dataset, consisting of 850 challenging images.

TABLE II

COMPARISON WITH STATE-OF-THE-ART SOD METHODS ON FIVE BENCHMARK DATASETS. BEST RESULTS ARE HIGHLIGHTED IN **BOLD**

Model	Pub.Year	DUTS-TE				ECSSD				HKU-IS				PASCAL-S				DUT-OMRON			
		$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$
VST[18]	ICCV'21	0.896	0.828	0.037	0.919	0.932	0.910	0.340	0.951	0.928	0.897	0.030	0.952	0.873	0.816	0.670	0.902	0.850	0.755	0.058	0.871
MSFNet[48]	ACMMM'21	0.877	0.841	0.034	0.931	0.914	0.916	0.033	0.954	0.908	0.903	0.027	0.959	0.849	0.830	0.063	0.902	0.832	0.757	0.050	0.772
PurNet[49]	TIP'21	0.892	0.842	0.036	0.924	0.930	0.920	0.031	0.953	0.924	0.910	0.026	0.957	0.860	0.819	0.063	0.901	0.842	0.756	0.055	0.871
DHQNet[50]	ICCV'21	0.892	0.860	0.032	0.933	0.927	0.928	0.029	0.957	0.923	0.919	0.025	0.960	0.859	0.833	0.060	0.905	0.840	0.770	0.050	0.877
DCNet[2]	TIP'21	0.892	0.840	0.035	0.925	0.928	0.920	0.031	0.953	0.922	0.905	0.027	0.956	0.861	0.820	0.062	0.900	0.845	0.760	0.051	0.875
ZoomNet[33]	CVPR'22	0.900	0.854	0.033	0.936	0.935	0.926	0.027	0.963	0.931	0.918	0.023	0.967	0.869	0.844	0.057	0.917	0.841	0.755	0.053	0.872
EDN[51]	TIP'22	0.892	0.845	0.035	0.908	0.927	0.918	0.032	0.929	0.924	0.908	0.026	0.956	0.864	0.833	0.061	0.870	0.849	0.770	0.049	0.879
PDRNet[52]	TCSVT'22	0.877	0.845	0.035	-	0.927	0.920	0.032	-	0.924	0.911	0.027	-	0.865	0.816	0.061	-	0.846	0.756	0.052	-
PoolNet+[53]	TPAMI'23	0.887	0.817	0.037	0.910	0.926	0.904	0.035	0.896	0.921	0.900	0.034	0.947	0.865	0.809	0.065	0.896	0.831	0.725	0.054	0.848
MENet[54]	CVPR'23	0.905	0.870	0.028	0.938	0.928	0.920	0.031	0.951	0.927	0.917	0.023	0.96	0.872	0.838	0.054	0.910	0.850	0.771	0.045	0.871
UDNet[55]	PR'23	0.902	-	0.032	0.918	0.932	-	0.031	0.931	0.924	-	0.027	0.856	0.862	-	0.059	0.865	-	-	-	-
PRNet[56]	TMM'23	0.879	0.811	0.039	0.910	0.917	0.895	0.330	0.946	0.913	0.885	0.032	0.956	0.853	0.808	0.067	0.891	0.829	0.723	0.056	0.866
TCRNet[57]	TCSVT'23	0.880	0.842	0.034	-	0.928	0.917	0.031	-	0.923	0.908	0.026	-	0.865	0.817	0.059	-	0.843	0.748	0.054	-
ICON-R[58]	TPAMI'23	0.889	0.837	0.037	0.924	0.929	0.918	0.032	0.954	0.920	0.902	0.029	0.953	0.861	0.818	0.064	0.899	0.844	0.761	0.057	0.876
DSRNet[21]	TCSVT'23	0.903	0.882	0.029	0.945	0.936	0.941	0.023	0.966	0.929	0.930	0.021	0.968	0.875	0.856	0.049	0.923	0.852	0.794	0.051	0.888
SelfReformer[19]	TMM'23	0.911	0.872	0.027	0.943	0.936	0.926	0.027	0.957	0.931	0.915	0.024	0.960	0.874	0.848	0.051	0.919	0.861	0.784	**0.043**	0.884
Ours		**0.922**	**0.899**	**0.023**	**0.956**	**0.946**	**0.943**	**0.021**	**0.969**	**0.940**	**0.932**	**0.020**	**0.970**	**0.885**	**0.862**	**0.048**	**0.926**	**0.873**	**0.810**	0.045	**0.903**

DUTS is a relatively large dataset with two subsets. DUTS-TR, containing 10553 images, is used for training, and DUTS-TE, containing 5019 images, is used for testing. For the COD task, we use the training subset of the CAMO and COD10K for training as done in many previous methods. We evaluated our model on four widely used public datasets: CAMO [40], CHAMELEON [41], COD10K [42], and NC4K [43]. Specifically, COD10K is currently the largest dataset for camouflage object detection, consisting of 10,000 images, including 5066 camouflage object images, 3000 background images, and 1934 non-camouflage object images. The CHAMELEON dataset is small, containing only 76 images, all sourced from the internet. The CAMO dataset comprises 1250 camouflage images, with 1000 used for training and the remaining 250 for testing. The NC4K dataset is the most recently released camouflage object dataset, including 4121 camouflage images, primarily used for evaluating the performance of camouflage object recognition models.

Similarly, we selected five widely used evaluation metrics in the field to assess our method, namely, MAE [44], $S-measure(S_m)$ [45], $Weighted\ F-measure(F_\beta^\omega)$ [46], and $E-measure(E_m)$ [47]. MAE evaluates the average pixel-wise difference between the predicted map (P) and the ground-truth map (G).

The MAE is calculated as follows:

$$MAE = \frac{1}{W \times H} \sum_{x=1}^{W} \sum_{y=1}^{H} |P(x, y) - G(x, y)| \tag{11}$$

$S-measure$ is primarily used to assess the structural similarity between the predicted map and the ground truth, calculated by the following formula:

$$S_m = ms_0 + (1 - m)s_r \tag{12}$$

where so and sr denote the object-aware and region-aware structural similarity and m is set to 0.5. $WeightedF-measure$ is an extension of F_m. By assigning different weights (ω) to different errors based on the specific location and neighborhood information.

$$F_\beta^\omega = \frac{(1 + \beta^2)Precision^\omega Recall^\omega}{\beta^2 Precision^\omega + Recall^\omega} \tag{13}$$

$E-measure$ combines the local pixel values with the image-level mean value in one term and can be computed as:

$$E_m = \frac{1}{W \times H} \sum_{x=1}^{W} \sum_{y=1}^{H} \Theta(m) \tag{14}$$

where m is the alignment matrix and $\Theta(m)$ indicates the enhanced alignment matrix.

C. Comparison With State-of-the-Art Methods

We compare our method with 16 state-of-the-art salient object detection methods: VST [18], MSFNet [70], PurNet [49], DHQNet [50], DCNet [2], ZoomNet [33], EDN [51], PDRNet [52], PoolNet+ [53], MENet [54], UDNet [55], PRNet [56], TCRNet [57], ICON [58], DSRNET [21], SelfReformer [19]. Meanwhile, our approach is compared with 20 state-of-the-art camouflaged object detection methods: PFNet [59], MGL [60], C2FNet [61], UGTR [62], UJSC [63], SINetV2 [24], BgNet [28], BGNet [1], BSANet [29], SegMaR [64], FDNet [65], ZoomNet [33], ICON [58], DGNet [30], BiRRNet [66], DTCNet [67], FEDER [68], MFFN [5], SARNet [69], MSCAF-Net [70]. The saliency maps and camouflaged maps are either provided by their authors or calculated using the released code. For a fair comparison, all 16 SOD methods we compared used the same training and test sets as ours. Similarly, all 20 COD methods we compared used the same training and test sets as ours.

1) Quantitative Comparison: We present quantitative comparison results on both SOD and COD datasets in Table II and Table III, respectively. It is evident from the tables that our method achieves optimal results across all datasets. Although our approach does not achieve the best MAE metric on the $DUT-OMRON$ dataset, the difference with the best result is minimal. Particularly noteworthy is the substantial reduction of 14.8% in the MAE metric compared to the second-best method on the $DUTS-TE$ dataset. In addition, Fig.5 and Fig.6 depict the Precision-Recall and F-measure curves, respectively, demonstrating a more detailed and comprehensive comparison between our method and others.

Furthermore, we compared our method with other popular approaches in terms of Floating-point operations ($FLOPs$) and the number of parameters ($Params$). As shown in

TABLE III

COMPARISON WITH STATE-OF-THE-ART COD METHODS ON FOUR BENCHMARK DATASETS. BEST RESULTS ARE HIGHLIGHTED IN **BOLD**

Model	Pub.Year	COD10K				CHAMELEON				NC4K				CAMO			
		$S_m \uparrow$	$F^\omega_\beta \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F^\omega_\beta \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F^\omega_\beta \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F^\omega_\beta \uparrow$	MAE $\downarrow$	$E_m \uparrow$
PFNet[59]	CVPR'21	0.800	0.660	0.040	0.877	0.882	0.810	0.033	0.931	0.829	0.745	0.053	0.887	0.782	0.695	0.085	0.841
MGL-R[60]	CVPR'21	0.814	0.666	0.035	0.851	0.893	0.812	0.031	0.917	0.833	0.739	0.053	0.867	0.775	0.673	0.088	0.812
C2FNet[61]	IJCAI'21	0.813	0.686	0.036	0.890	0.888	0.828	0.032	0.935	0.838	0.762	0.049	0.897	0.796	0.719	0.080	0.854
UGTR[62]	ICCV'21	0.818	0.667	0.035	0.853	0.888	0.796	0.031	0.911	0.839	0.747	0.052	0.874	0.785	0.686	0.086	0.823
UJSC[63]	CVPR'21	0.809	0.684	0.035	0.884	0.891	0.833	0.030	0.945	0.842	0.771	0.047	0.898	0.800	0.728	0.073	0.859
SINetV2[24]	TPAMI'22	0.815	0.680	0.037	0.887	0.888	0.816	0.030	0.942	0.847	0.770	0.048	0.903	0.820	0.743	0.070	0.882
BgNet[28]	KBS'22	0.826	0.703	0.034	0.898	0.894	0.823	0.029	0.943	0.855	0.784	0.045	0.907	0.832	0.762	0.065	0.884
BGNet[1]	IJCAI'22	0.831	0.722	0.033	0.901	0.901	0.850	0.027	0.943	0.851	0.788	0.044	0.907	0.812	0.749	0.073	0.870
BSANet[29]	AAAI'22	0.818	0.699	0.034	0.891	0.895	0.841	0.027	0.946	0.841	0.771	0.048	0.897	0.794	0.717	0.079	0.851
SegMaR[64]	CVPR'22	0.833	0.724	0.034	0.899	0.906	0.860	0.025	0.951	0.841	0.781	0.046	0.896	0.815	0.753	0.071	0.874
FDNet[65]	CVPR'22	0.840	0.729	0.030	0.919	0.897	0.835	0.027	0.950	0.834	0.750	0.052	0.893	0.841	0.775	0.063	0.895
ZoomNet[33]	CVPR'22	0.838	0.729	0.029	0.888	0.902	0.845	0.023	0.943	0.853	0.784	0.043	0.896	0.820	0.752	0.066	0.877
ICON[58]	TPAMI'23	0.818	0.688	0.033	0.904	0.854	0.763	0.037	0.923	0.858	0.785	0.041	0.922	0.840	0.769	0.058	0.904
DGNet[30]	MIR'23	0.822	0.693	0.033	0.896	0.890	0.816	0.029	0.938	0.857	0.784	0.042	0.911	0.839	0.769	0.057	0.901
Bi-RRNet[66]	PR'23	0.840	0.746	0.026	0.912	0.896	0.852	0.022	0.960	0.861	0.810	0.038	0.917	0.843	0.802	0.054	0.909
DTC-Net[67]	TMM'23	0.79	0.616	0.041	0.821	0.876	0.773	0.039	0.879	-	-	-	-	0.778	0.667	0.084	0.804
FEDER[68]	CVPR'23	0.822	0.716	0.032	0.900	0.887	0.834	0.030	0.946	0.847	0.789	0.044	0.907	0.802	0.738	0.071	0.867
MFFN[5]	WACV'23	0.846	0.745	0.028	0.917	0.905	0.852	0.021	0.963	0.856	0.791	0.042	0.915	-	-	-	-
SARNet[69]	TCSVT'23	0.864	0.777	0.024	0.931	0.912	0.871	0.021	0.957	0.886	0.842	0.032	0.937	0.868	0.828	0.047	0.927
MSCAF-Net[70]	TCSVT'23	0.865	0.775	0.024	0.927	0.912	0.865	0.022	0.958	0.887	0.839	0.032	0.935	0.873	0.828	0.046	0.929
Ours		**0.877**	**0.799**	**0.022**	**0.936**	**0.922**	**0.882**	**0.019**	**0.969**	**0.894**	**0.850**	**0.030**	**0.938**	**0.879**	**0.839**	**0.045**	**0.930**

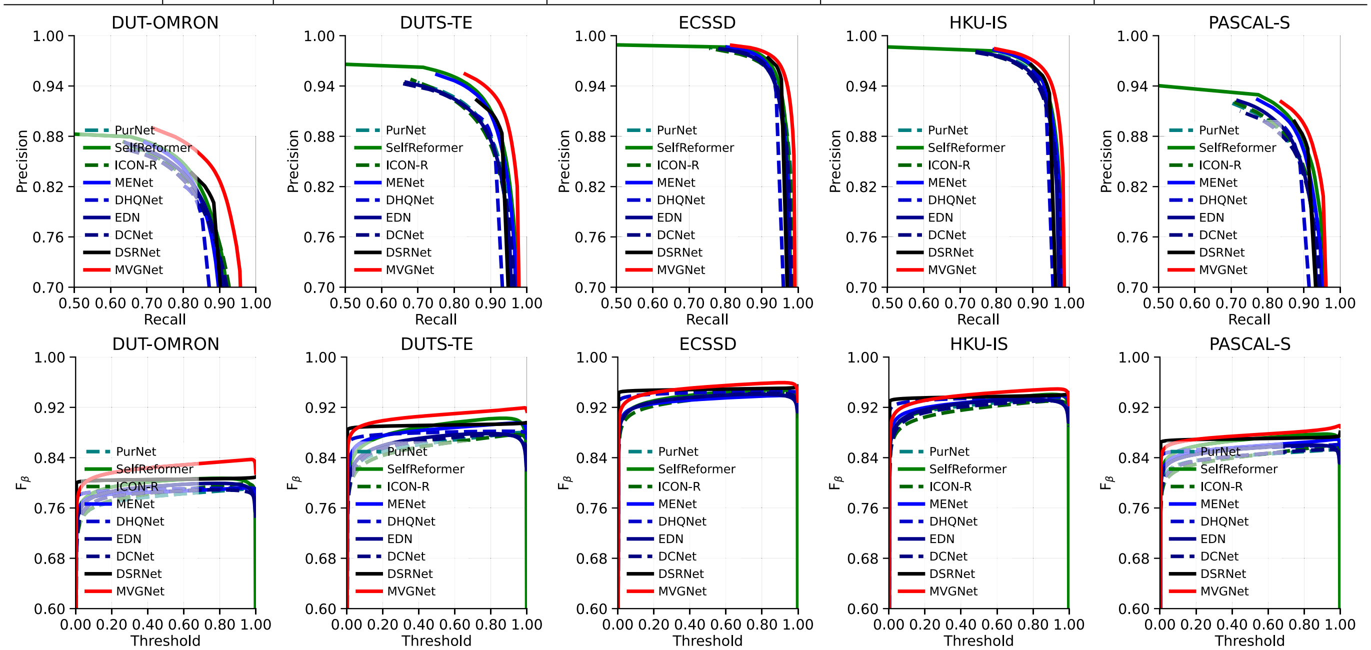

Fig. 5. Our method is compared with relevant approaches on publicly available datasets in the field of salient object detection. The first row depicts the precision-recall (PR) curves, and the second row presents the F-measure (F_m) curves, with our results highlighted in red.

TABLE IV

OUR METHOD IS COMPARED WITH OTHER SOD METHODS IN TERMS OF FLOPS AND THE NUMBER OF PARAMETERS

Methods / sets	VST[18]	MSFNet[48]	PurNet[49]	DHQNet[50]	DCNet[2]	ZoomNet[33]	EDN[51]	PDRNet[52]	PRNet[56]	TCRNet[57]	ICON[58]	DSRNet[21]	MVGNet(Ours)
Input size	224×224	256×256	320×320	224×224	224×224	384×384	384×384	320×320	320×320	320×320	384×384	224×224	352×352
FLOPs(G)	23.24	24.48	47.74	10.44	22.37	218.43	20.42	24.60	35.85	20.12	24.96	19.18	29.00
Params(M)	44.09	33.35	35.53	59.47	37.96	33.16	42.85	47.70	-	57.82	33.04	115.84	56.11

TABLE V

OUR METHOD IS COMPARED WITH OTHER COD METHODS IN TERMS OF FLOPS AND THE NUMBER OF PARAMETERS

Methods / sets	PFNet[59]	MGL[60]	C2FNet[61]	UGTR[62]	UJSC[63]	SINetv2[24]	ZoomNet[33]	ICON[58]	DGNet[30]	Bi-RRNet[66]	SARNet[69]	MSCAF-Net[70]	MVGNet(Ours)
Input size	416×416	416×416	384×384	473×473	352×352	352×352	384×384	384×384	352×352	448×448	384×384	352×352	352×352
FLOPs(G)	26.50	378.20	21.23	123.30	36.62	23.30	218.43	24.96	-	12.65	23.15	-	29.00
Params(M)	46.17	73.73	26.36	37.54	55.94	45.53	33.16	33.04	21.36	14.95	47.20	29.68	56.11

Table IV, we present a comparison of the $Params$t and $FLOPs$ between MVGNet and 12 SOD models. Table V shows the comparison of the $Params$ and $FLOPs$ between MVGNet and 12 COD models. For a fair comparison, all the

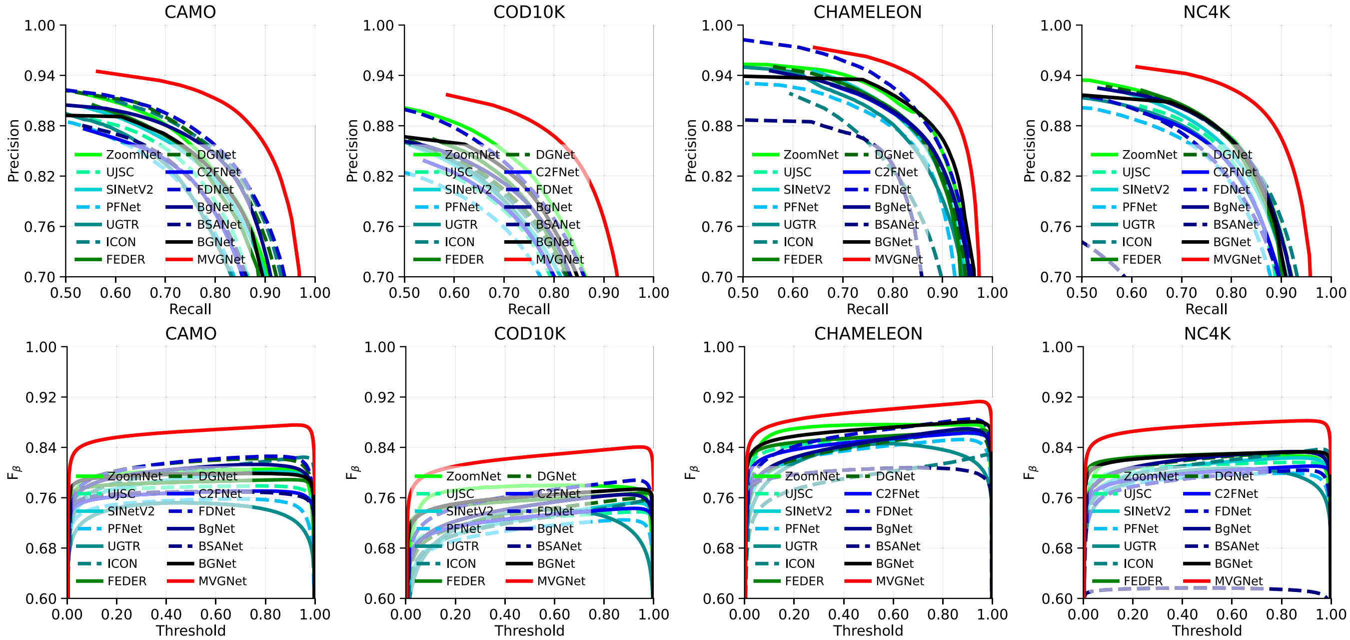

Fig. 6. Our method is compared with relevant approaches on publicly available datasets in the field of camouflaged object detection. The first row depicts the precision-recall (PR) curves, and the second row presents the F-measure (F_m) curves, with our results highlighted in red.

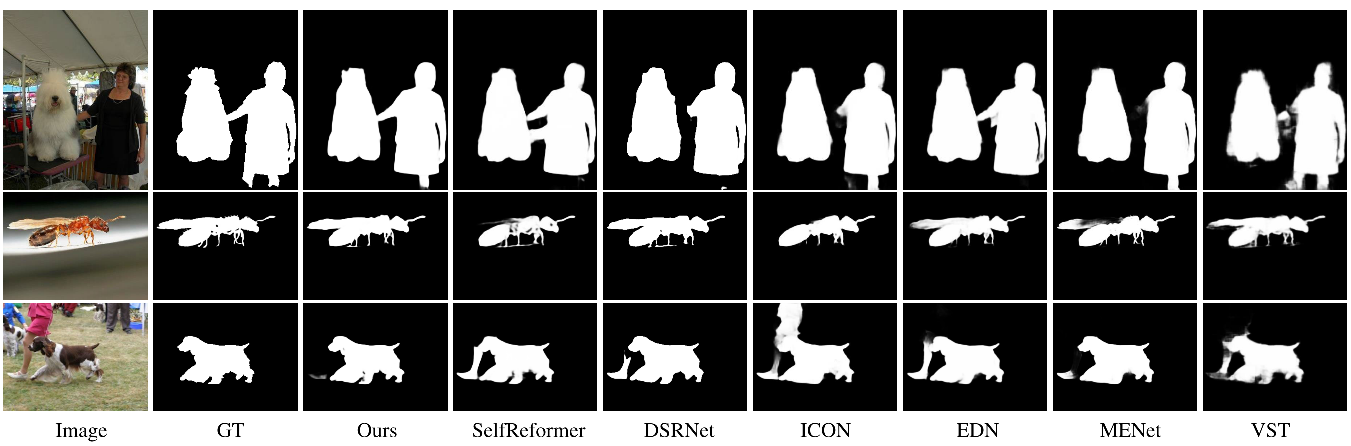

Fig. 7. Visual comparison with other salient object detection methods.

results we compared are from the publicly available results in the authors' published papers. It is worth noting that our method uses a CNN branch to extract multi-view information, effectively controlling the increase in computational complexity while achieving accurate prediction results. Compared to ZoomNet [33], which also employs a multi-view strategy, our method has a significant advantage in terms of FLOPs. Additionally, compared to DSRNet [21], which adopts a dual-branch network architecture, our method has an advantage in terms of the number of parameters. These benefits are attributed to our network design strategy.

2) Qualitative Comparison: To more intuitively demonstrate the superior performance of our approach, we present the comparison of prediction results for some representative images, as shown in Fig.7 and 8. The first row of Fig.7 and the first row of Fig.8 display an image with two objects, posing a challenge to the model's discriminative ability. It can be observed that our method not only identifies the objects well but also accurately recognizes the details between the two targets, while other methods exhibit blurriness or errors in the connection between the two targets. In the second row of Fig.7 and the second row of Fig.8, we present objects with fine structures. Compared to other methods, our approach excels in recovering detailed information about objects. However, there are still some instances of ambiguity. We attribute this to the multi-view strategy, which, while enhancing global perceptual capabilities, somewhat diminishes the emphasis on detailed information. We will provide a detailed elaboration on this point in the subsequent section dedicated to the analysis of failure cases. In the third row of Fig.7 and Fig.8, we showcase images with complex scenes, containing one or more targets. This tests the network's perceptual ability for target saliency. Thanks to our attention exploration module, our model can effectively identify the salient object in the center of the image, as well as the camouflaged object hidden in the background.

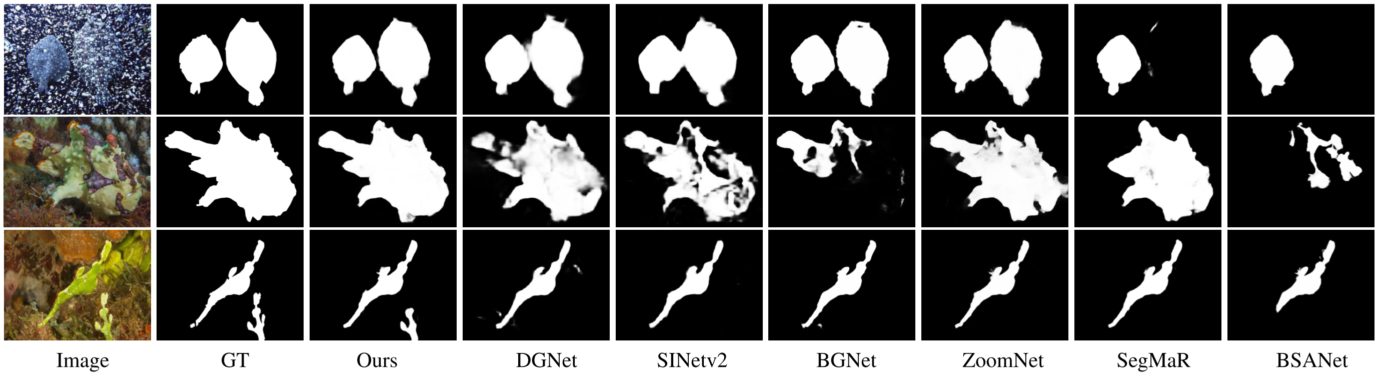

Fig. 8. Visual comparison with other camouflaged object detection methods.

TABLE VI
THE RESULTS OF ABLATION EXPERIMENTS ON FOUR PUBLICLY AVAILABLE CAMOUFLAGED OBJECT DATASETS. BEST RESULTS ARE HIGHLIGHTED IN **BOLD**

MFM	AEM	Size	Param	FLOPs	COD10K				CHAMELEON				NC4K				CAMO			
					$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$
	✓	352x352	53.33M	26.40G	0.831	0.761	0.025	0.924	0.901	0.874	0.019	0.947	0.872	0.820	0.032	0.917	0.867	0.800	0.047	0.923
✓		352x352	55.94M	28.76G	0.853	0.776	0.023	0.929	0.916	0.880	0.019	0.962	0.880	0.836	0.032	0.930	0.870	0.817	0.046	0.922
✓	✓	352x352	56.11M	29.00G	**0.877**	**0.799**	**0.022**	**0.936**	**0.922**	**0.882**	**0.019**	**0.969**	**0.894**	**0.850**	**0.030**	**0.938**	**0.879**	**0.839**	**0.045**	**0.930**

TABLE VII
THE RESULTS OF ABLATION EXPERIMENTS ON FIVE PUBLICLY AVAILABLE SALIENT OBJECT DATASETS. BEST RESULTS ARE HIGHLIGHTED IN **BOLD**

MFM	AEM	DUTS-TE				ECSSD				HKU-IS				PASCAL-S				DUT-OMRON			
		$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$
	✓	0.906	0.867	0.025	0.939	0.911	0.877	0.024	0.931	0.930	0.921	0.021	0.962	0.880	0.839	0.049	0.917	0.851	0.803	0.048	0.892
✓		0.910	0.869	0.024	0.945	0.912	0.908	0.025	0.932	0.931	0.926	0.020	0.959	0.881	0.854	0.049	0.915	0.854	0.790	0.046	0.876
✓	✓	**0.922**	**0.899**	**0.023**	**0.956**	**0.946**	**0.943**	**0.021**	**0.969**	**0.940**	**0.932**	**0.020**	**0.970**	**0.885**	**0.862**	**0.048**	**0.926**	**0.873**	**0.810**	**0.045**	**0.903**

D. Ablation Study

In this section, We conducted ablation analysis on the core components of the model using nine publicly available datasets. Table VI presents the ablation experiment results of MVGNet on four public datasets in the COD domain, while Table VII shows the ablation experiment results of MVGNet on five public datasets in the SOD domain.

1) Effectiveness of the CNN Encoder and MFM: We removed the CNN branch from the network to assess its impact on the model's performance. The results, as shown in the first row of Table VI, reveal a significant reduction in computational complexity after removing the CNN branch, aligning with our expectations. This is because the component is responsible for extracting multi-view features, leading to higher computational demands. However, compared to other networks employing a similar multi-view strategy, our computational load remains within a relatively small range. Moreover, analyzing network performance, the introduction of this component has substantially enhanced the model's capabilities, demonstrating significant growth across various evaluation metrics.

Furthermore, to control the consumption of computational resources, we did not employ a backbone network for multi-view information extraction, as in other works. Instead, we independently designed a CNN-based Multi-View Encoder(MVE), combined with a multi-view fusion module, which significantly reduces computational resource consumption while maintaining accuracy. As shown in Table VIII, to demonstrate the parameter advantage of this module, we conducted experimental comparisons with a Transformer-based multi-view encoder. We tested it on the DUTS-TE dataset because it is the largest SOD dataset and can comprehensively reflect the impact of changes in each component on model performance.

TABLE VIII
COMPARISON RESULTS BETWEEN CNN-BASED AND TRANSFORMER-BASED MULTI-VIEW ENCODERS

MVE	Size	Param	FLOPs	DUTS-TE			
				$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE $\downarrow$	$E_m \uparrow$
CNN	352x352	56.11M	29.00G	0.922	0.899	0.023	0.956
Transformer	352x352	91.87M	96.41G	0.923	0.896	0.023	0.958

From the experimental results, it can be seen that using a CNN-based encoder reduces the model parameters by 38.92% and FLOPs by 69.92% compared to using a Transformer-based encoder, while achieving comparable segmentation performance. The main reason for this is that the multi-view encoder processes information from three perspectives, so if this module is designed to be complex, it will result in a significant increase in computational resource consumption. Additionally, we designed the MFM to fuse multi-view information, so the

TABLE IX

COMPARISON RESULTS WITH OTHER POLYP SEGMENTATION METHODS ON FOUR BENCHMARK DATASETS. BEST RESULTS ARE HIGHLIGHTED IN **BOLD**

Model	CVC-ColonDB				CVC-ClinicDB				ETIS				Kvasir			
	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE ↓	Dice ↑	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE ↓	Dice ↑	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE ↓	Dice ↑	$S_m \uparrow$	$F_\beta^\omega \uparrow$	MAE ↓	Dice ↑
U-Net[71]	0.711	0.498	0.061	0.519	0.889	0.811	0.019	0.824	0.682	0.366	0.036	0.406	0.858	0.794	0.055	0.821
U-Net++[72]	0.691	0.467	0.064	0.490	0.872	0.785	0.022	0.797	0.681	0.390	0.035	0.413	0.862	0.808	0.048	0.824
ZoomNet[33]	0.773	0.612	0.045	0.619	0.915	0.863	0.015	0.861	0.758	0.539	0.020	0.562	0.904	0.868	0.031	0.872
PraNet[22]	0.82	0.699	0.043	0.716	**0.935**	0.896	**0.009**	0.902	0.791	0.600	0.031	0.63	0.915	0.885	0.030	0.901
TPRNet[73]	0.829	0.632	0.047	0.735	0.932	0.814	0.014	0.909	0.812	0.550	0.032	0.703	0.907	0.815	0.038	0.885
SINetV2[24]	0.852	0.750	0.032	0.770	0.927	**0.901**	0.014	0.905	0.832	0.669	0.019	0.703	0.909	0.881	0.027	0.892
Ours	**0.882**	**0.806**	**0.028**	**0.902**	0.933	0.896	0.014	**0.930**	**0.876**	**0.752**	**0.018**	**0.959**	**0.921**	**0.898**	**0.026**	**0.924**

Fig. 9. Several representative failure cases of our methd.

encoder does not require excessive redundant features, and good performance can be achieved using CNN.

2) Effectiveness of the AEM: We replaced the AEM component with a simple convolutional module to assess its impact on the model's performance, as shown in the second row of Table VI and Table VII. Specifically, we utilized a multiplication strategy and concatenation operation along with convolution to hierarchically fuse multi-view features with Transformer features. Upsampling operations were interleaved during the fusion process to gradually generate high-resolution prediction maps. It is evident that after the replacement, there is a significant decrease in all evaluation metrics on both COD and SOD datasets. This further underscores the effective ability of our AEM component to distinguish between salient and camouflaged objects.

3) Failure Cases and Analyses: Despite our model achieving excellent evaluation results across all datasets, it still encounters challenges in recognizing targets with fine structures. As illustrated in Fig. 9, we present several failure cases where our model tends to miss or blur predictions in certain detailed areas. We speculate that this phenomenon arises due to the enhancement of perceptual capabilities through multi-view information, which, in turn, overlooks some detailed information.

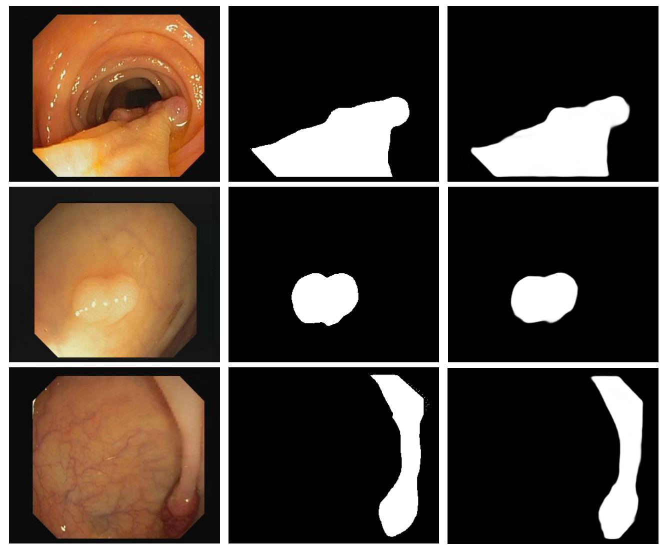

Fig. 10. Visual cases of polyp segmentation.

V. DOWNSTREAM APPLICATIONS

Thanks to our design, the model has the ability to perceive the inherent characteristics of objects, namely saliency and camouflage. Therefore, in other segmentation domains that also involve saliency or camouflage, our method demonstrates good generalizability and robustness. To further demonstrate the generality of our approach, we apply it to other segmentation tasks, including polyp segmentation and defect detection. The relevant experimental results are presented in this Section.

A. Polyp Segmentation

The polyp segmentation task aims to assist doctors in detecting polyps from colonoscopy images, which holds significant importance for surgical procedures. We applied our network to this domain and conducted experiments on four challenging publicly available datasets, namely Kvasir [74], CVC-ClinicCB [75], ETIS [76], and CVC-ColonDB [77]. We utilized a portion of the data from Kvasir and

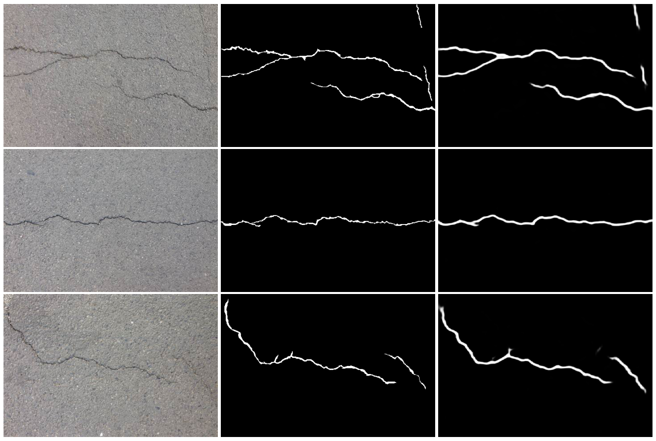

Fig. 11. Visual cases of road defect detection.

CVC-ClinicCB as the training set, a common practice in this field. The test results on the four datasets are presented in Table IX. We conducted performance comparisons using four widely adopted evaluation metrics—Sm,F_β^ω,MAE, and $Dice$—against other methods, including U-Net [71], U-Net++ [72], ZoomNet [33], PraNet [22], TPRNet [73], and SINetV2 [24]. Fig. 10 presents some visualized results.

B. Defect Detection

Road crack detection is a type of defect detection that holds significant importance for the maintenance and repair of roads. We further trained our model on a road crack detection dataset(CrackForest [78]), and Fig. 11 presents some visualized results.

VI. Conclusion

We propose a novel multi-view guided network for salient and camouflaged object detection. Our focus lies in leveraging commonalities and distinguishing differences between salient and camouflaged objects. We design a CNN-based encoder and fusion module for multi-view information extraction, effectively enhancing the recognition accuracy for both, while rigorously controlling the increase in computational cost. Additionally, an attention exploration module is devised to enhance perception from different perspectives. We explore color transformation relationships to identify camouflaged objects and spatial positional relationships to determine salient objects. The results from both aspects are then fused to generate precise prediction images.

As our method is a general approach applicable to both COD and SOD, we conduct detailed experiments on nine publicly available datasets in these domains. The experimental results demonstrate competitive performance compared to 35 state-of-the-art methods in both domains. In addition, we have validated the strong generality of our model on relevant visual tasks, including polyp segmentation and defect detection. In the future, we plan to further improve the network's recognition accuracy, with a specific emphasis on enhancing its capability to identify fine-grained details.

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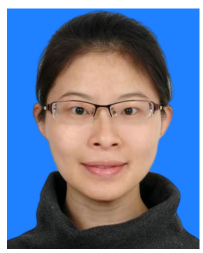

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